EE2080 : Lecture 2 : 07 August 2026

RECAP: 1) Random Vectors : $\underline{X} = [X_1, \ldots, X_m]$ X_i random variable defined w.r.t $(\Omega, \mathcal{F}, \mathbb{P})$

Mean: $\mu_X = [\mathbb{E}[X_1] \ldots \mathbb{E}[X_m]]^T$

Autocorrelation : $\mathbb{E}[XX^T]_{m\times m}$ with $\mathbb{E}[XX^T]_{i,j} = \mathbb{E}[X_iX_j]$; Autocovariance.

Crosscorrelation: For 2 Random vectors $\underline{X}_m, \underline{Y}_n$ on same $(\Omega, \mathcal{F}, \mathbb{P})$: $\mathbb{E}[\underline{X}\underline{Y}^T]_{m\times n}$

2) Random Processes : $\{X_t : t \in T\}$ a collection of random variables defined on $(\Omega, \mathcal{F}, \mathbb{P})$

T: index set : Discrete & Continuous Parameter Processes.

Homework from Lecture 1

Example 2 : $X_1 \sim \text{unif}[0,1]$, $X_2 = X_1 + W$, $W \sim \text{unif}[0,1]$ $X_1 \perp W$

Let $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$. Find $\mathbb{E}[X]$, $\mathbb{E}[XX^T]$ and $\text{cov}(X)$.

$$\mathbb{E}[X] = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} \quad \text{Cov}(X) = \begin{bmatrix} 1/12 & 1/12 \\ 1/12 & 1/6 \end{bmatrix} \quad \mathbb{E}[XX^T] = \begin{bmatrix} 1/3 & 7/12 \\ 7/12 & 7/6 \end{bmatrix}$$

Example 3 : Consider $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$, where X_1, X_2 are random variables defined on the same probability space with joint distribution

$$f_{X_1,X_2}(x_1,x_2) = \begin{cases} 6x_2 , & 0 \le x_1 \le 1 , \ 0 \le x_2 \le x_1 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{E}[X]$ and $\mathbb{E}[XX^T]$

$$\mathbb{E}[X] = \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix} \quad \mathbb{E}[XX^T] = \begin{bmatrix} 3/5 & 2/5 \\ 2/5 & 3/10 \end{bmatrix}$$

RANDOM PROCESS : Definition

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A random process is a collection of random variables $\{X_t, t \in T\}$, indexed by the elements of an index set T, all defined w.r.t $\mathcal{F}$.

- When T is finite, say $T = \{1, 2, \cdots, d\}$, we have a random vector $X = [X_1 \cdots X_d]^T$
- If T is a countably infinite set, $\{X_t, t \in T\}$ is a discrete parameter process
- If T is uncountably infinite $\{X_t, t \in T\}$ is a continuous parameter process.
- Often, T has the interpretation of time.
- Random processes are also called Stochastic Processes (from a Greek word for "guess")

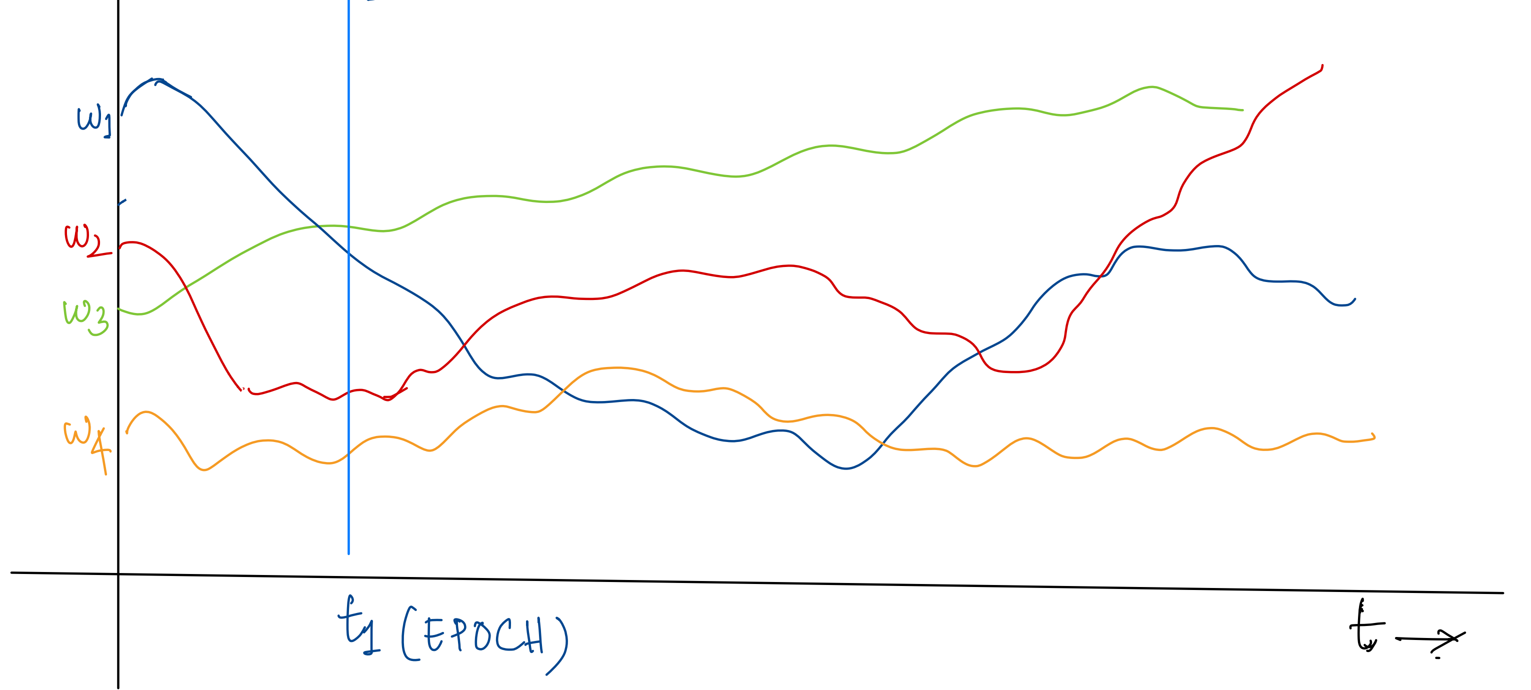
Interpretations of a Random Process $\{X_t, t \in T\}$

$X_t : \Omega \longrightarrow \mathbb{R}$ is a random variable for each $t \in T$

$X_{\cdot}(\omega) : T \longrightarrow \mathbb{R}$ is a sample path of the process for each $\omega \in \Omega$.

$X : T \times \Omega \longrightarrow \mathbb{R}$ $\quad \underbrace{X_t(\omega) \text{ or } X(t, \omega)}_{\text{as a function of two variables}}$ is a real number for each $\omega \in \Omega$, $t \in T$.

Typically, T is considered as $\mathbb{Z}, \mathbb{N}, \mathbb{R}$ & $\mathbb{R}^+$

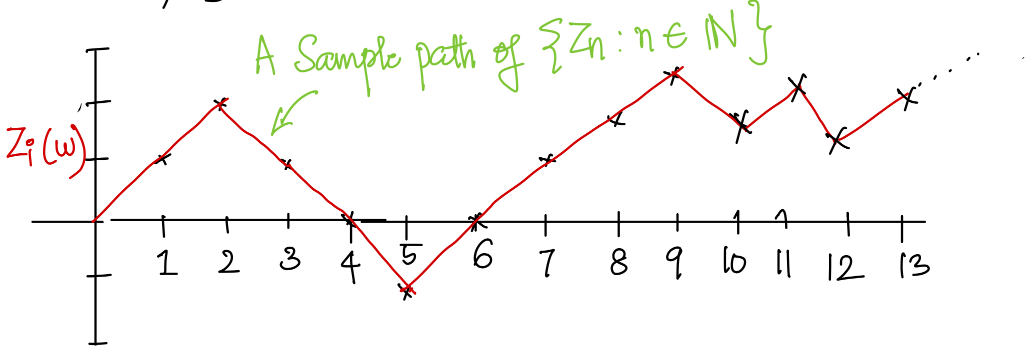


Example: Random Walk.

Qn: When is an infinite seq. of RVs independent?

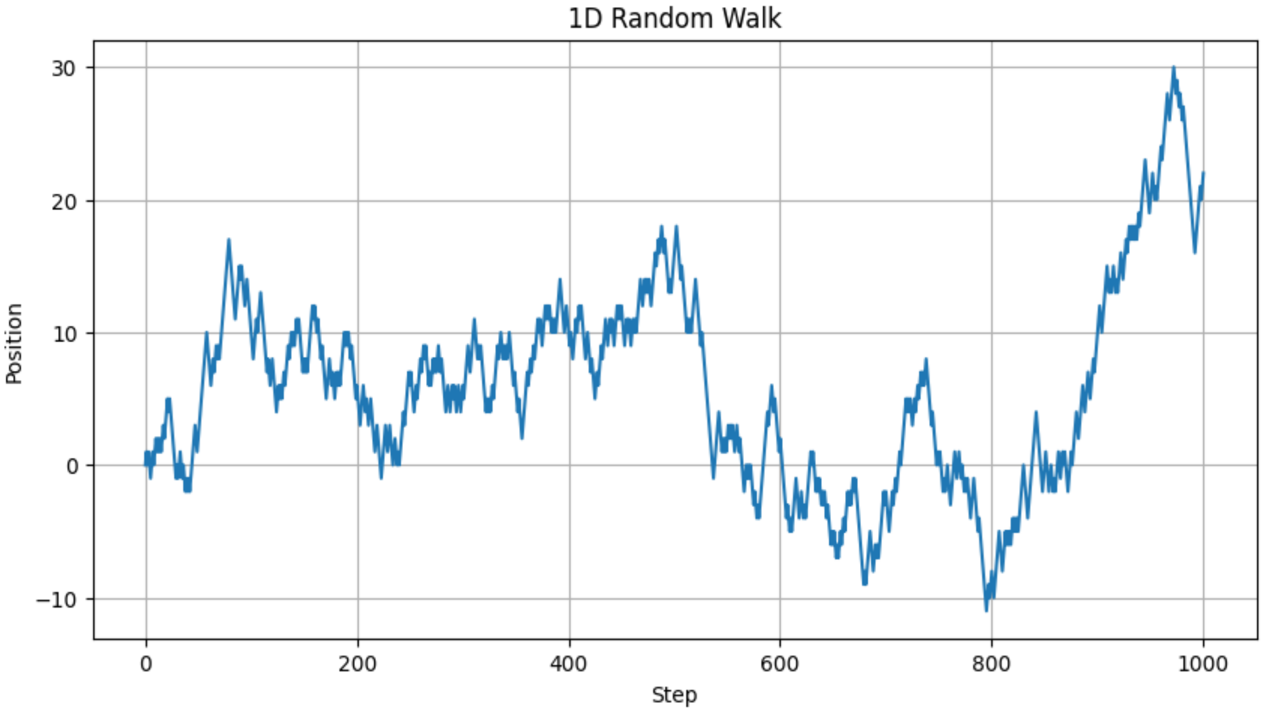
Consider a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

$X_1, X_2, X_3, \dots$ defined w.r.t $\mathcal{F}$, i.i.d with $\mathbb{P}(X_i = 1) = \frac{1}{2}$ and $\mathbb{P}(X_i = -1) = \frac{1}{2}$.



$$Z_n = \sum_{i=1}^{n} X_i \quad , \; n \in \mathbb{N}$$

$Z_1, Z_2, \dots$ is a random walk.



Finite Dimensional Distributions (FDDs)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Defn: Let $\{X_t, t \in T\}$ be a random process defined w.r.t $\mathcal{F}$.
For any $m \in \mathbb{N}$ and a finite set $\{t_1, t_2, \dots, t_m\} \subset T$,
$$F_{t_1, t_2, \dots t_m}(x_1, x_2, \dots, x_m) = \mathbb{P}(X_{t_1} \le x_1, X_{t_2} \le x_2, \dots, X_{t_m} \le x_m), \quad (x_1, \dots x_m) \in \mathbb{R}^m$$
is called a finite dimensional distribution of the process $\{X_t, t \in T\}$

The collection of all such finite dimensional distributions (i.e, $\forall m \in \mathbb{N}$, and all finite sets $\{t_1, t_2, \dots, t_m\} \subset T$) is called the set of FDDs for the process $\{X_t, t \in T\}$

Remark: FDDs cannot be an arbitrary collection of joint CDFs

For example: If (X,Y) is a random vector, then the joint distribution $F_{X,Y}(x,y)$ and the individual distributions $F_X(x)$ and $F_Y(y)$ cannot be arbitrary

They need to satisfy: $F_X(x) = F_{X,Y}(x,\infty)$ and $F_Y(y) = F_{X,Y}(\infty, y)$.

i.e., the distributions should be consistent.

Consistency of FDDs

Defn: The set of FDDs of the process $\{X_t, t \in T\}$ are said to be consistent if for any $m, n \in \mathbb{N}$, and subsets $T_m \subset T_n \subset T$ with $|T_m| = m$ and $|T_n| = n$,

$$F_{T_m}(x_1, \dots x_m) = F_{T_n}(\infty, \dots, \infty, x_1, \infty, \dots, \infty, x_2, \dots\dots, \infty, \dots, \infty, x_m, \infty \dots \infty)$$

where the dimensions in T_n that are not in T_m are shown as ∞ in the RHS.

Properties of Random Processes: Mean, Autocorrelation and Autocovariance

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $\{X_t : t \in T\}$ be a random process defined w.r.t $\mathcal{F}$.

The mean of the process $\{X_t : t \in T\}$ is a function $\mu_X : T \to [-\infty, +\infty]$ defined as

$$\boxed{\mu_X(t) = \mathbb{E}[X_t]}, \quad t \in T.$$

ENSEMBLE MEAN

The autocorrelation and autocovariance of the process $\{X_t : t \in T\}$ are functions $R_X, C_X : T \times T \longrightarrow [-\infty, +\infty]$ defined as

$$\boxed{R_X(t,s) = \mathbb{E}[X_t X_s]}, \quad \boxed{C_X(t,s) = \mathrm{Cov}(X_t, X_s)}, \quad t,s \in T.$$

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A second order random process is a random process $\{X_t : t \in T\}$ such that $\mathbb{E}[X_t^2] < \infty \ \forall t \in T$.

The mean, correlation and covariance functions of a second order random process are all well-defined and finite.

Consider a random process $\{X_t : t \in T\}$ defined over a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

- If X_t is a discrete random variable for each $t \in T$, then the n^{th} order PMF of the RP $\{X_t : t \in T\}$ is defined by $\boxed{P_{t_1, t_2, \ldots, t_n}(x_1, \ldots, x_n) = \mathbb{P}(\{X_{t_1} = x_1, \ldots, X_{t_n} = x_n\})}$

- Similarly if $X_{t_1}, X_{t_2}, \ldots, X_{t_n}$ are jointly continuous random variables for any distinct $t_1, t_2, \ldots, t_n \in T$, then $\{X_t : t \in T\}$ has an n^{th} order pdf s.t for each $(t_1, t_2, \ldots, t_n)$ fixed, $\boxed{f_{t_1, \ldots, t_n}(x_1, \ldots, x_n)}$ is the joint pdf of $X_{t_1}, X_{t_2}, \ldots, X_{t_n}$.

Example: Let A and B be iid $\mathcal{N}(0,1)$ random variables.

Suppose $X_t = A + Bt + t^2 \ \forall t \in \mathbb{R}$.

Describe $\mu_X(t)$, $R_X(s,t)$, $C_X(s,t)$, the first and second order pdfs of $\{X_t : t \in \mathbb{R}\}$

$$\mu_X(t) = \mathbb{E}[X_t] = \mathbb{E}[A + Bt + t^2] = t^2$$

$$R_X(s,t) = \mathbb{E}[X_s X_t] = 1 + st + s^2 t^2$$

$$C_X(s,t) = R_X(s,t) - \mu_X(s)\mu_X(t) = 1 + st$$

For a fixed t, X_t is a linear combination of 2 $\mathcal{N}(0,1)$ RVs $\therefore$ a Gaussian RV with mean $\mu_X(t) = t^2$ and variance $C_X(t,t) = 1 + t^2$ $\therefore X_t \sim \mathcal{N}(t^2, 1 + t^2)$

$\therefore$ First order pdf: $f_{t_1}(x_1) = \dfrac{1}{\sqrt{2\pi(1+t^2)}} \exp\left(-\dfrac{(x-t^2)^2}{2(1+t^2)}\right)$

For distinct s and t, $Cov(X_s, X_t) = \begin{bmatrix} 1+s^2 & 1+st \\ 1+st & 1+t^2 \end{bmatrix}$ $\quad |Cov(X_s, X_{st})| = (s-t)^2 \neq 0$

$\therefore \{X_t, t \in \mathbb{R}\}$ has a second order pdf.

$$f_{s,t} = \frac{1}{2\pi|s-t|} \exp\left\{\frac{-1}{2}\begin{pmatrix} x-s^2 \\ y-t^2 \end{pmatrix}^T \begin{bmatrix} 1+s^2 & 1+st \\ 1+st & 1+t^2 \end{bmatrix}^{-1} \begin{bmatrix} x-s^2 \\ y-t^2 \end{bmatrix}\right)$$

Example: Consider $\{W_n\}_{n=1}^{\infty}$, $W_{n_s} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$ Homework!

Define the process $\{X_n\}_{n=1}^{\infty}$ as $X_n = W_n + W_{n-1}$, $X_1 = W_1$

Find the mean, autocorrelation of the process $\{X_n\}_{n\geq 1}$

Is $\{X_n\}_{n\geq 1}$ an IID process? What can you say about the N^{th} order PDFs of $\{X_n\}_{n\geq 1}$?

Processes with independent increments:

A random process $\{X_t\}_{t\in T}$ is said to have independent increments if $\forall\, t_1, t_2, \ldots, t_N \in T$, s.t. $0 \leq t_1 < t_2 \cdots < t_N$, $N \in \mathbb{N}$, the increments $X_{t_2} - X_{t_1}, X_{t_3} - X_{t_2}, \ldots, X_{t_N} - X_{t_{N-1}}$ are mutually independent

The increment $X_{t_2} - X_{t_1}$ represents the change in process between times t_1 and t_2 $(t_2 > t_1)$

Independent increments $\Rightarrow$ knowing the evolution of the process in one interval does not provide any information about what happens in another disjoint interval.

Example: Brownian motion (Wiener Process)

$\{W_t\}_{t\geq 0}$ with i) $W_t = 0$, ii) $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $0 \leq s < t$.

iii) W_t has independent increments., iv) W_t is almost surely continuous.

$F_{t_1}(x_1) = F_{t_2}(x_1) \;\forall t_1, t_2$

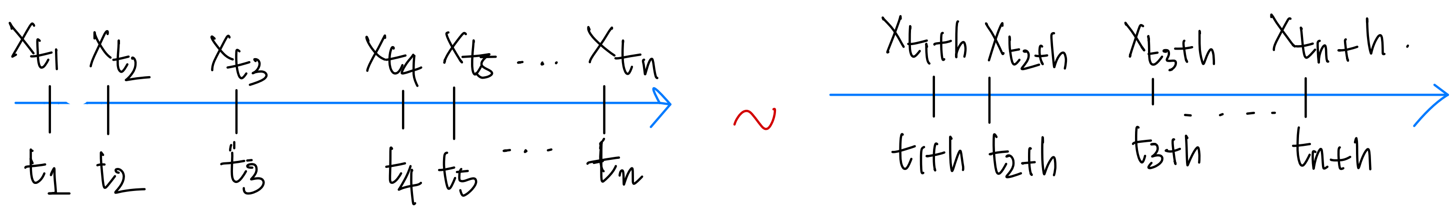
$F_{t_1,t_2}(x_1,x_2) = F_{t_1+h, t_2+h}(x_1,x_2)$
$\forall t_1, t_2, h \in \mathbb{R}$

Stationary Process:

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $\{X_t, t \in \mathbb{R}\}$ be a random process defined w.r.t $\mathcal{F}$

Defn: The random process $\{X_t, t \in \mathbb{R}\}$ is said to be (strictly) stationary if all FDDs are translation-invariant, i.e., for any $n \in \mathbb{N}$, $(t_1, \dots, t_n) \in \mathbb{R}^n$ and $h \in \mathbb{R}$, $F_{t_1,\dots,t_n}(x_1,\dots,x_n) = F_{t_1+h,\dots,t_n+h}(x_1,\dots,x_n)$, $\forall (x_1,\dots,x_n) \in \mathbb{R}^n$



N^{th} Order Stationary Process:

A random process $\{X_t, t \in \mathbb{R}\}$ defined w.r.t the prob. space $(\Omega, \mathcal{F}, \mathbb{P})$ is said to be

- first-order stationary if $F_t(x) = F_{t+h}(x) \quad \forall t, h \in \mathbb{R}, x \in \mathbb{R}$.
- Second-order stationary if $F_{t_1,t_2}(x_1,x_2) = F_{t_1+h,t_2+h}(x_1,x_2) \;\forall t_1, t_2, h \in \mathbb{R}, x_1, x_2 \in \mathbb{R}$
  and $F_t(x) = F_{t+h}(x), \forall t, h \in \mathbb{R}, x \in \mathbb{R}$

  In particular, if we take $h = -t_1$, we have $F_{t_1,t_2}(x_1,x_2) = F_{0,t_2-t_1}(x_1,x_2)$
  The second order joint distribution only depends on the shift $t_2 - t_1$ and not on the values of t_2 and t_1.

  $R_X(t_1,t_2) = \mathbb{E}[X_{t_1} X_{t_2}]$ will be a function of only $t_2 - t_1$ and not the actual values of t_2 and t_1 $\therefore$ For a 2^{nd} order stationary process, the autocorrelation fn is denoted as $R_X(t_2 - t_1)$ or $R_X(\Delta)$ where Δ denotes the shift, $\Delta \in \mathbb{R}$

- N^{th} order stationary if $F_{t_1,\dots,t_n}(x_1,\dots,x_n) = F_{t_1+h,\dots,t_n+h}(x_1,\dots,x_n)$, for all $t_1,\dots,t_n \in \mathbb{R}$ and $x_1,\dots,x_n \in \mathbb{R}$, $\forall n \in \{1, 2, \dots, N\}$.

Wide Sense Stationary (WSS) Process

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Let $\{X_t : t \in \mathbb{R}^+\}$ be a random process defined w.r.t $\mathcal{F}$.

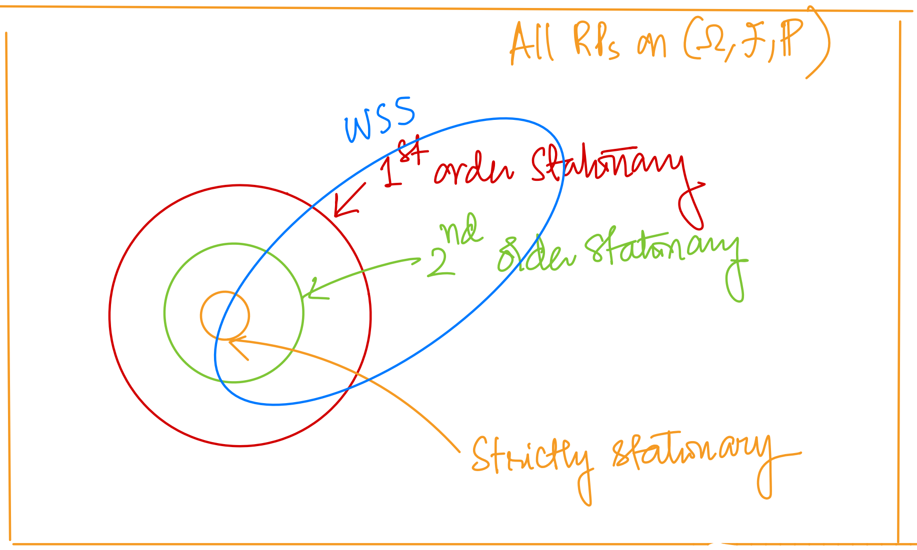
Defn: $\{X_t : t \in \mathbb{R}\}$ is said to be wide-sense stationary (OR weakly stationary) if for all $t_1, t_2 \in \mathbb{R}^+$ and $h \in \mathbb{R}^+$

1) $\mu_X(t_1) = \mu_X(t_2) \Rightarrow \mu_X$ is a constant
2) $C_X(t_1, t_2) = C_X(t_1 + h, t_2 + h) \Rightarrow \underbrace{C_X(t_2 - t_1)}$ $\underbrace{R_X(t_2 - t_1)}$
3) $\mathbb{E}[X_t^2] < \infty, \ \forall t \in \mathbb{R}$

Prove 1) Every stationary process with finite variance is WSS.
2) Every second order stationary process with finite variance is WSS.

Homework

Qn: Where will WSS processes lie? (Containment relation)



$\{X_t : t \in \mathbb{R}\}$
$(\Omega, \mathcal{F}, \mathbb{P})$

Example 4 Flip a fair coin once. If the outcome is H, then $\{X_n\}_{n \geq 1}$ is defined as $X_n = 1 \ \forall n$. If the outcome is T, $X_n = W_n \ \forall n \geq 1$ where $W_n \overset{iid}{\sim} \text{Ber}(p)$. Is the process $\{X_n\}_{n \geq 1}$

a) WSS? b) SSS? Are X_ns independent?

Example: 1) Stationary but not WSS. : IID Cauchy Process.

Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$. Consider $\{X_t : t \in T\}$ where X_t iid Cauchy(0,1) $\forall t \in T$

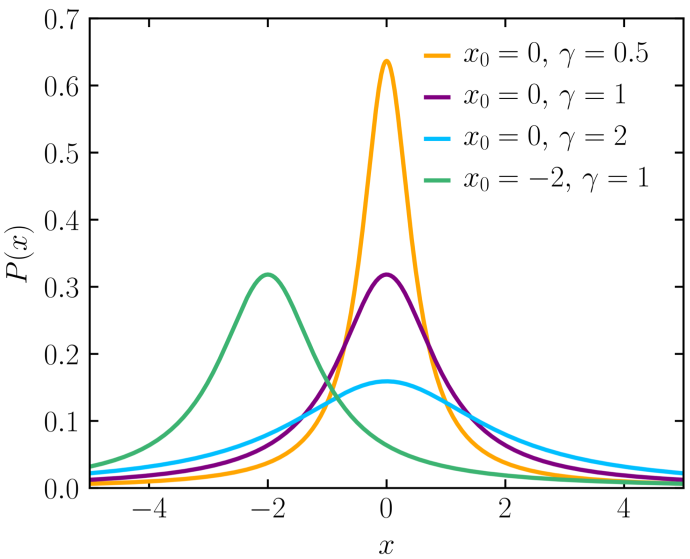
Cauchy (Lorentzian) Distribution :

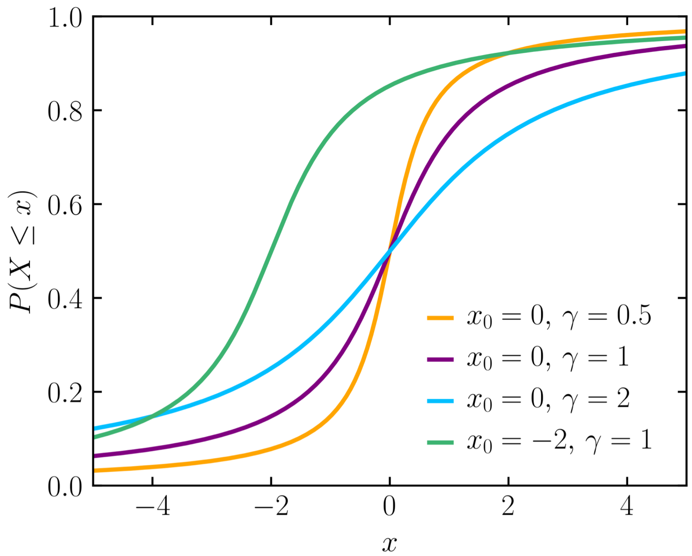
A continuous RV. $X \sim$ Cauchy(0,1) has a pdf : $f_X(x) = \dfrac{1}{\pi(1+x^2)}$, $x \in (-\infty, \infty)$

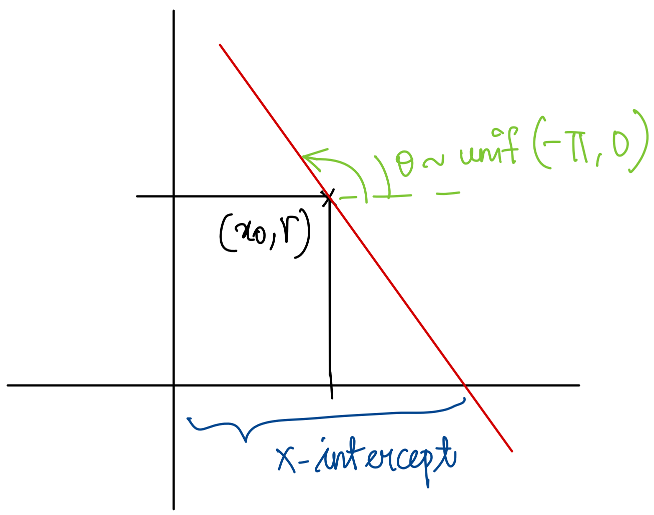
In general: if $X \sim$ Cauchy(x_0, γ), $f_X(x) = \dfrac{1}{\pi\gamma\left[1 + \dfrac{(x-x_0)^2}{\gamma^2}\right]}$, $x \in (-\infty, \infty)$

CDF : $F_X(x) = \dfrac{1}{\pi} \tan^{-1}\left(\dfrac{x - x_0}{\gamma}\right) + \dfrac{1}{2}$

x_0 : location
γ : scale.







$f_X(x, x_0, \gamma)$ is the distribution of the x-intercept of a line passing through (x_0, γ) with a uniformly distributed angle.

Generating a sample from a Cauchy(0,1) distribution:

Let u be a sample from unif (0,1).

$x = \tan\left(\pi(u - \tfrac{1}{2})\right)$ is a sample from Cauchy(0,1)

Cauchy distribution has no mean, variance or higher moments defined..

$\therefore$ IID Cauchy process is stationary but not WSS!!

Example : 2) WSS but not stationary

$\{X_n : n \in \mathbb{N} \cup \{0\}\}$ on some $(\Omega, \mathcal{F}, \mathbb{P})$.

$X_{2n} \overset{iid}{\sim} \text{unif}\{-1, 0, 1\}$ $\quad\quad \mathbb{E}[X_{2n}] = 0$

$X_{2n+1} = |X_{2n}| - 2/3$ $\quad\quad \mathbb{E}[X_{2n+1}] = 2/3 * 1/3 - 1/3 * 2/3 = 0$

$\mathbb{E}[X_s X_t]$: $s \neq t$.

$$X_{2n+1} X_{2n} = \begin{cases} -1/3 & \text{w.p. } 1/3 \\ 0 & \text{w.p } 1/3 \\ 1/3 & \text{w.p. } 1/3 . \end{cases}$$

$\mathbb{E}[X_{2n+1} \cdot X_{2n}] = 0$

For other values of s and t $X_s \perp X_t$ $\therefore \mathbb{E}[X_s X_t] = \mathbb{E}[X_s]\mathbb{E}[X_t] = 0$

$\mathbb{E}[X_{2n}^2] = 1 \times \frac{1}{3} + 0 \times \frac{1}{3} + 1 \times 1/3 = \underline{\underline{2/3}}$

$\mathbb{E}[X_{2n+1}^2] = (2/3)^2 \times 1/3 + (1/3)^2 \times 2/3 = 4/9 \times \frac{1}{3} + \frac{1}{9} \times 2/3$

$= 6/27 = \underline{\underline{2/9}}$

$\therefore \{X_n : n \geq 0\}$ is a WSS process.

Is it stationary?

$$F_{X_{2n}}(x) = \begin{cases} 0, & x < -1 \\ 1/3, & -1 \leq x < 0 \\ 2/3, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$F_{X_{2n+1}}(x) = \begin{cases} 0, & x < -2/3 \\ 1/3, & -2/3 \leq x < 1/3 \\ 1, & x \geq 1/3 . \end{cases}$$

$\therefore$ Not Stationary!!

Example 3 : $\{X_n : n \in \mathbb{N}\}$, $\quad X_n = \sin n\Theta$, $\quad \Theta \sim \text{unif}(0, 2\pi)$

Homework

Is this process a) WSS ?

b) SSS ?